Writing Exercises

223. Explain how subtraction and addition are related.

224. How does knowing addition facts help you to subtract numbers?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use subtraction notation.			
model subtraction of whole numbers.			
subtract whole numbers.			
translate word phrases to math notation.			
subtract whole numbers in applications.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

1.4 Multiply Whole Numbers

Learning Objectives

By the end of this section, you will be able to:

- Use multiplication notation
- Model multiplication of whole numbers
- Multiply whole numbers
- Translate word phrases to math notation
- Multiply whole numbers in applications

☑ **BE PREPARED** 1.5 Before you get started, take this readiness quiz.

Add: $1{,}683 + 479$.
If you missed this problem, review Example 1.21.

☑ **BE PREPARED** 1.6 Subtract: $605 - 321$.
If you missed this problem, review Example 1.33.

Use Multiplication Notation

Suppose you were asked to count all these pennies shown in Figure 1.11.



Figure 1.11

Would you count the pennies individually? Or would you count the number of pennies in each row and add that number 3 times.

$$8 + 8 + 8$$

Multiplication is a way to represent repeated addition. So instead of adding 8 three times, we could write a multiplication expression.

$$3 \times 8$$

We call each number being multiplied a factor and the result the **product**. We read 3×8 as *three times eight*, and the result as *the product of three and eight*.

There are several symbols that represent multiplication. These include the symbol $\times$ as well as the dot, $\cdot$, and

parentheses ( ).

Operation Symbols for Multiplication

To describe multiplication, we can use symbols and words.

Operation	Notation	Expression	Read as	Result
Multiplication	$\times$ $\cdot$ ( )	3×8 $3 \cdot 8$ $3(8)$	three times eight	the product of 3 and 8

EXAMPLE 1.39

Translate from math notation to words:

ⓐ 7×6 ⓑ $12 \cdot 14$ ⓒ $6(13)$

Solution

ⓐ We read this as *seven times six* and the result is *the product of seven and six.*

ⓑ We read this as *twelve times fourteen* and the result is *the product of twelve and fourteen.*

ⓒ We read this as *six times thirteen* and the result is *the product of six and thirteen.*

TRY IT 1.77 Translate from math notation to words:

ⓐ 8×7 ⓑ $18 \cdot 11$

TRY IT 1.78 Translate from math notation to words:

ⓐ $(13)(7)$ ⓑ $5(16)$

Model Multiplication of Whole Numbers

There are many ways to model multiplication. Unlike in the previous sections where we used base-10 blocks, here we will use counters to help us understand the meaning of multiplication. A counter is any object that can be used for counting. We will use round blue counters.

EXAMPLE 1.40

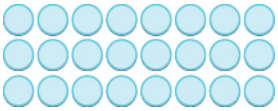
Model: 3×8.

Solution

To model the product 3×8, we'll start with a row of 8 counters.



The other factor is 3, so we'll make 3 rows of 8 counters.



Now we can count the result. There are 24 counters in all.

$3 \times 8 = 24$

If you look at the counters sideways, you'll see that we could have also made 8 rows of 3 counters. The product would have been the same. We'll get back to this idea later.

TRY IT 1.79 Model each multiplication: 4×6.

TRY IT 1.80 Model each multiplication: 5×7.

Multiply Whole Numbers

In order to multiply without using models, you need to know all the one digit multiplication facts. Make sure you know them fluently before proceeding in this section.

Table 1.4 shows the multiplication facts. Each box shows the product of the number down the left column and the number across the top row. If you are unsure about a product, model it. It is important that you memorize any number facts you do not already know so you will be ready to multiply larger numbers.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	10	12	14	16	18
3	0	3	6	9	12	15	18	21	24	27
4	0	4	8	12	16	20	24	28	32	36
5	0	5	10	15	20	25	30	35	40	45
6	0	6	12	18	24	30	36	42	48	54
7	0	7	14	21	28	35	42	49	56	63
8	0	8	16	24	32	40	48	56	64	72
9	0	9	18	27	36	45	54	63	72	81

Table 1.4

What happens when you multiply a number by zero? You can see that the product of any number and zero is zero. This is called the Multiplication Property of Zero.

Multiplication Property of Zero

The product of any number and 0 is 0.

$$a \cdot 0 = 0$$
$$0 \cdot a = 0$$

EXAMPLE 1.41

Multiply:

ⓐ $0 \cdot 11$ ⓑ $(42)0$

Solution

(a)	$0 \cdot 11$
The product of any number and zero is zero.	0
(b)	$(42)0$
Multiplying by zero results in zero.	0

TRY IT 1.81 Find each product:

(a) $0 \cdot 19$ (b) $(39)0$

TRY IT 1.82 Find each product:

(a) $0 \cdot 24$ (b) $(57)0$

What happens when you multiply a number by one? Multiplying a number by one does not change its value. We call this fact the Identity Property of Multiplication, and 1 is called the multiplicative identity.

Identity Property of Multiplication

The product of any number and 1 is the number.

$$1 \cdot a = a$$
$$a \cdot 1 = a$$

EXAMPLE 1.42

Multiply:

(a) $(11)1$ (b) $1 \cdot 42$

Solution

(a)	$(11)1$
The product of any number and one is the number.	11
(b)	$1 \cdot 42$
Multiplying by one does not change the value.	42

TRY IT 1.83 Find each product:

(a) $(19)1$ (b) $1 \cdot 39$

TRY IT 1.84 Find each product:

(a) $(24)(1)$ (b) 1×57

Earlier in this chapter, we learned that the Commutative Property of Addition states that changing the order of addition

does not change the sum. We saw that $8 + 9 = 17$ is the same as $9 + 8 = 17$.

Is this also true for multiplication? Let's look at a few pairs of factors.

$$4 \cdot 7 = 28 \qquad 7 \cdot 4 = 28$$
$$9 \cdot 7 = 63 \qquad 7 \cdot 9 = 63$$
$$8 \cdot 9 = 72 \qquad 9 \cdot 8 = 72$$

When the order of the factors is reversed, the product does not change. This is called the Commutative Property of Multiplication.

Commutative Property of Multiplication

Changing the order of the factors does not change their product.

$$a \cdot b = b \cdot a$$

EXAMPLE 1.43

Multiply:

ⓐ $8 \cdot 7$ ⓑ $7 \cdot 8$

Solution

ⓐ	$8 \cdot 7$
Multiply.	56
ⓑ	$7 \cdot 8$
Multiply.	56

Changing the order of the factors does not change the product.

TRY IT 1.85 Multiply:

ⓐ $9 \cdot 6$ ⓑ $6 \cdot 9$

TRY IT 1.86 Multiply:

ⓐ $8 \cdot 6$ ⓑ $6 \cdot 8$

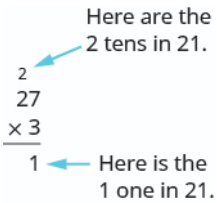
To multiply numbers with more than one digit, it is usually easier to write the numbers vertically in columns just as we did for addition and subtraction.

$$\begin{array}{r} 27 \\ \times 3 \\ \hline \end{array}$$

We start by multiplying 3 by 7.

$$3 \times 7 = 21$$

We write the 1 in the ones place of the product. We carry the 2 tens by writing 2 above the tens place.



Then we multiply the 3 by the 2, and add the 2 above the tens place to the product. So $3 \times 2 = 6$, and $6 + 2 = 8$. Write the 8 in the tens place of the product.

$$\begin{array}{r} {\scriptstyle 2} \\ 27 \\ \times\ 3 \\ \hline 81 \end{array}$$ ← This comes from 3×2 plus the 2 we carried.

The product is 81.

When we multiply two numbers with a different number of digits, it's usually easier to write the smaller number on the bottom. You could write it the other way, too, but this way is easier to work with.

EXAMPLE 1.44

Multiply: $15 \cdot 4$.

Solution

Write the numbers so the digits 5 and 4 line up vertically.	$\begin{array}{r} 15 \\ \times\ 4 \\ \hline \end{array}$
Multiply 4 by the digit in the ones place of 15. $4 \cdot 5 = 20$.	
Write 0 in the ones place of the product and carry the 2 tens.	$\begin{array}{r} {\scriptstyle 2} \\ 15 \\ \times\ 4 \\ \hline 0 \end{array}$
Multiply 4 by the digit in the tens place of 15. $4 \cdot 1 = 4$. Add the 2 tens we carried. $4 + 2 = 6$.	
Write the 6 in the tens place of the product.	$\begin{array}{r} {\scriptstyle 2} \\ 15 \\ \times\ 4 \\ \hline 60 \end{array}$

TRY IT 1.87 Multiply: $64 \cdot 8$.

TRY IT 1.88 Multiply: $57 \cdot 6$.

EXAMPLE 1.45

Multiply: $286 \cdot 5$.

Solution

Write the numbers so the digits 5 and 6 line up vertically.	$\begin{array}{r} 286 \\ \times\ 5 \\ \hline \end{array}$
Multiply 5 by the digit in the ones place of 286. $5 \cdot 6 = 30$.	

Write the 0 in the ones place of the product and carry the 3 to the tens place.Multiply 5 by the digit in the tens place of 286. $5 \cdot 8 = 40$.	$\begin{array}{r} {\scriptstyle 3} \\ 286 \\ \times\ 5 \\ \hline 0 \end{array}$
Add the 3 tens we carried to get $40 + 3 = 43$. Write the 3 in the tens place of the product and carry the 4 to the hundreds place.	$\begin{array}{r} {\scriptstyle 4\,3} \\ 286 \\ \times\ 5 \\ \hline 30 \end{array}$
Multiply 5 by the digit in the hundreds place of 286. $5 \cdot 2 = 10$. Add the 4 hundreds we carried to get $10 + 4 = 14$. Write the 4 in the hundreds place of the product and the 1 to the thousands place.	$\begin{array}{r} {\scriptstyle 4\,3} \\ 286 \\ \times\ 5 \\ \hline 1{,}430 \end{array}$

TRY IT 1.89 Multiply: $347 \cdot 5$.

TRY IT 1.90 Multiply: $462 \cdot 7$.

When we multiply by a number with two or more digits, we multiply by each of the digits separately, working from right to left. Each separate product of the digits is called a partial product. When we write partial products, we must make sure to line up the place values.

HOW TO

Multiply two whole numbers to find the product.

Step 1. Write the numbers so each place value lines up vertically.
Step 2. Multiply the digits in each place value.
- Work from right to left, starting with the ones place in the bottom number.
- Multiply the ones digit of the bottom number by the ones digit in the top number, then by the tens digit, and so on.
- If a product in a place value is more than 9, carry to the next place value.
- Write the partial products, lining up the digits in the place values with the numbers above.
- Repeat for the tens place in the bottom number, the hundreds place, and so on.
- Insert a zero as a placeholder with each additional partial product.

Step 3. Add the partial products.

EXAMPLE 1.46

Multiply: $62\,(87)$.

Solution

Write the numbers so each place lines up vertically.	$\begin{array}{r} 62 \\ \times\ 87 \\ \hline \end{array}$
Start by multiplying 7 by 62. Multiply 7 by the digit in the ones place of 62. $7 \cdot 2 = 14$. Write the 4 in the ones place of the product and carry the 1 to the tens place.	$\begin{array}{r} {\scriptstyle 1} \\ 62 \\ \times\ 87 \\ \hline 4 \end{array}$
Multiply 7 by the digit in the tens place of 62. $7 \cdot 6 = 42$. Add the 1 ten we carried. $42 + 1 = 43$. Write the 3 in the tens place of the product and the 4 in the hundreds place.	$\begin{array}{r} {\scriptstyle 1} \\ 62 \\ \times\ 87 \\ \hline 434 \end{array}$
The first partial product is 434.	
Now, write a 0 under the 4 in the ones place of the next partial product as a placeholder since we now multiply the digit in the tens place of 87 by 62. Multiply 8 by the digit in the ones place of 62. $8 \cdot 2 = 16$. Write the 6 in the next place of the product, which is the tens place. Carry the 1 to the tens place.	$\begin{array}{r} {\scriptstyle 1} \\ 62 \\ \times\ 87 \\ \hline 434 \\ 60 \end{array}$
Multiply 8 by 6, the digit in the tens place of 62, then add the 1 ten we carried to get 49. Write the 9 in the hundreds place of the product and the 4 in the thousands place.	$\begin{array}{r} {\scriptstyle 1} \\ 62 \\ \times\ 87 \\ \hline 434 \\ 4960 \end{array}$
The second partial product is 4960. Add the partial products.	$\begin{array}{r} {\scriptstyle 1} \\ 62 \\ \times\ 87 \\ \hline 434 \\ 4960 \\ \hline 5394 \end{array}$

The product is 5,394.

TRY IT 1.91 Multiply: $43(78)$.

TRY IT 1.92 Multiply: $64(59)$.

EXAMPLE 1.47

Multiply:

(a) $47 \cdot 10$ (b) $47 \cdot 100$.

Solution

(a)	$47 \cdot 10.$	$\begin{array}{r} 47 \\ \times 10 \\ \hline 00 \\ 470 \\ \hline 470 \end{array}$
(b)	$47 \cdot 100$	$\begin{array}{r} 47 \\ \times 100 \\ \hline 00 \\ 000 \\ 4700 \\ \hline 4{,}700 \end{array}$

When we multiplied 47 times 10, the product was 470. Notice that 10 has one zero, and we put one zero after 47 to get the product. When we multiplied 47 times 100, the product was $4{,}700$. Notice that 100 has two zeros and we put two zeros after 47 to get the product.

Do you see the pattern? If we multiplied 47 times $10{,}000$, which has four zeros, we would put four zeros after 47 to get the product $470{,}000$.

TRY IT 1.93 Multiply:

(a) $54 \cdot 10$ (b) $54 \cdot 100.$

TRY IT 1.94 Multiply:

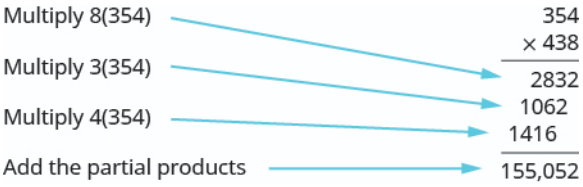
(a) $75 \cdot 10$ (b) $75 \cdot 100.$

EXAMPLE 1.48

Multiply: $(354)(438)$.

Solution

There are three digits in the factors so there will be 3 partial products. We do not have to write the 0 as a placeholder as long as we write each partial product in the correct place.



TRY IT 1.95 Multiply: $(265)(483)$.

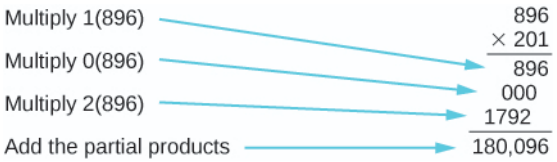
TRY IT 1.96 Multiply: $(823)(794)$.

EXAMPLE 1.49

Multiply: $(896)201$.

Solution

There should be 3 partial products. The second partial product will be the result of multiplying 896 by 0.



Notice that the second partial product of all zeros doesn't really affect the result. We can place a zero as a placeholder in the tens place and then proceed directly to multiplying by the 2 in the hundreds place, as shown.

Multiply by 10, but insert only one zero as a placeholder in the tens place. Multiply by 200, putting the 2 from the $2 \cdot 6 = 12$ in the hundreds place.

$$\begin{array}{r} 896 \\ \times 201 \\ \hline 896 \\ 17920 \\ \hline 180{,}096 \end{array}$$

TRY IT 1.97 Multiply: (718)509.

TRY IT 1.98 Multiply: (627)804.

When there are three or more factors, we multiply the first two and then multiply their product by the next factor. For example:

to multiply	$8 \cdot 3 \cdot 2$
first multiply $8 \cdot 3$	$24 \cdot 2$
then multiply $24 \cdot 2$.	48

Translate Word Phrases to Math Notation

Earlier in this section, we translated math notation into words. Now we'll reverse the process and translate word phrases into math notation. Some of the words that indicate multiplication are given in Table 1.5.

Operation	Word Phrase	Example	Expression
Multiplication	times product twice	3 times 8 the product of 3 and 8 twice 4	$3 \times 8, 3 \cdot 8, (3)(8),$ $(3)8,$ or $3(8)$ $2 \cdot 4$

Table 1.5

EXAMPLE 1.50

Translate and simplify: the product of 12 and 27.

Solution

The word *product* tells us to multiply. The words *of* 12 *and* 27 tell us the two factors.

	the product of 12 and 27
Translate.	$12 \cdot 27$
Multiply.	324

TRY IT 1.99 Translate and simplify: the product of 13 and 28.

TRY IT 1.100 Translate and simplify: the product of 47 and 14.

EXAMPLE 1.51

Translate and simplify: twice two hundred eleven.

Solution

The word *twice* tells us to multiply by 2.

	twice two hundred eleven
Translate.	2(211)
Multiply.	422

TRY IT 1.101 Translate and simplify: twice one hundred sixty-seven.

TRY IT 1.102 Translate and simplify: twice two hundred fifty-eight.

Multiply Whole Numbers in Applications

We will use the same strategy we used previously to solve applications of multiplication. First, we need to determine what we are looking for. Then we write a phrase that gives the information to find it. We then translate the phrase into math notation and simplify to get the answer. Finally, we write a sentence to answer the question.

EXAMPLE 1.52

Humberto bought 4 sheets of stamps. Each sheet had 20 stamps. How many stamps did Humberto buy?

Solution

We are asked to find the total number of stamps.

Write a phrase for the total.	the product of 4 and 20
Translate to math notation.	$4 \cdot 20$

Multiply.	$\begin{array}{r} 20 \\ \times\, 4 \\ \hline 80 \end{array}$
Write a sentence to answer the question.	Humberto bought 80 stamps.

TRY IT 1.103 Valia donated water for the snack bar at her son's baseball game. She brought 6 cases of water bottles. Each case had 24 water bottles. How many water bottles did Valia donate?

TRY IT 1.104 Vanessa brought 8 packs of hot dogs to a family reunion. Each pack has 10 hot dogs. How many hot dogs did Vanessa bring?

EXAMPLE 1.53

When Rena cooks rice, she uses twice as much water as rice. How much water does she need to cook 4 cups of rice?

Solution

We are asked to find how much water Rena needs.

Write as a phrase.	twice as much as 4 cups
Translate to math notation.	$2 \cdot 4$
Multiply to simplify.	8
Write a sentence to answer the question.	Rena needs 8 cups of water for 4 cups of rice.

TRY IT 1.105 Erin is planning her flower garden. She wants to plant twice as many dahlias as sunflowers. If she plants 14 sunflowers, how many dahlias does she need?

TRY IT 1.106 A college choir has twice as many women as men. There are 18 men in the choir. How many women are in the choir?

EXAMPLE 1.54

Van is planning to build a patio. He will have 8 rows of tiles, with 14 tiles in each row. How many tiles does he need for the patio?

Solution

We are asked to find the total number of tiles.

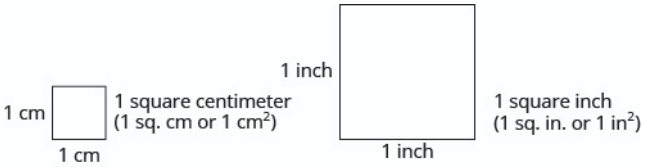
Write a phrase.	the product of 8 and 14
Translate to math notation.	$8 \cdot 14$

Multiply to simplify.	$\begin{array}{r} {}^{3}14 \\ \times 8 \\ \hline 112 \end{array}$
Write a sentence to answer the question.	Van needs 112 tiles for his patio.

TRY IT 1.107 Jane is tiling her living room floor. She will need 16 rows of tile, with 20 tiles in each row. How many tiles does she need for the living room floor?

TRY IT 1.108 Yousef is putting shingles on his garage roof. He will need 24 rows of shingles, with 45 shingles in each row. How many shingles does he need for the garage roof?

If we want to know the size of a wall that needs to be painted or a floor that needs to be carpeted, we will need to find its **area**. The area is a measure of the amount of surface that is covered by the shape. Area is measured in square units. We often use square inches, square feet, square centimeters, or square miles to measure area. A square centimeter is a square that is one centimeter (cm.) on a side. A square inch is a square that is one inch on each side, and so on.



For a rectangular figure, the area is the product of the length and the width. Figure 1.12 shows a rectangular rug with a length of 2 feet and a width of 3 feet. Each square is 1 foot wide by 1 foot long, or 1 square foot. The rug is made of 6 squares. The area of the rug is 6 square feet.

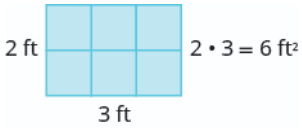


Figure 1.12 The area of a rectangle is the product of its length and its width, or 6 square feet.

EXAMPLE 1.55

Jen's kitchen ceiling is a rectangle that measures 9 feet long by 12 feet wide. What is the area of Jen's kitchen ceiling?

Solution

We are asked to find the area of the kitchen ceiling.

Write a phrase for the area.	the product of 9 and 12
Translate to math notation.	$9 \cdot 12$

Multiply.	$\begin{array}{r} {}^{1}12 \\ \times 9 \\ \hline 108 \end{array}$
Answer with a sentence.	The area of Jen's kitchen ceiling is 108 square feet.

TRY IT 1.109 Zoila bought a rectangular rug. The rug is 8 feet long by 5 feet wide. What is the area of the rug?

TRY IT 1.110 Rene's driveway is a rectangle 45 feet long by 20 feet wide. What is the area of the driveway?

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Multiplying Whole Numbers (https://www.openstax.org/l/24multwhlnum)
Multiplication with Partial Products (https://www.openstax.org/l/24multpartprod)
Example of Multiplying by Whole Numbers (https://www.openstax.org/l/24examplemultnm)

SECTION 1.4 EXERCISES

Practice Makes Perfect

Use Multiplication Notation

In the following exercises, translate from math notation to words.

225. 4×7 **226.** 8×6 **227.** $5 \cdot 12$

228. $3 \cdot 9$ **229.** $(10)(25)$ **230.** $(20)(15)$

231. $42(33)$ **232.** $39(64)$

Model Multiplication of Whole Numbers

In the following exercises, model the multiplication.

233. 3×6 **234.** 4×5 **235.** 5×9

236. 3×9

Multiply Whole Numbers

In the following exercises, fill in the missing values in each chart.

237.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0		0	0	0	0		0
1	0	1	2	3			6	7	8	
2		2	4	6	8		12			18
3	0		6		12	15		21		27
4	0	4			16	20		28	32	
5	0	5	10	15			30		40	
6	0	6	12		24			42		54
7			14	21		35			56	63
8	0	8		24			48		64	
9	0	9	18		36	45			72	

238.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0		0		0	0
1	0	1	2		4	5		7		9
2	0		4		8	10		14	16	
3		3		9			18		24	
4	0	4	8	12			24	28		36
5	0	5		15	20		30	35	40	
6			12	18			36	42		54
7	0	7		21		35			56	63
8	0	8	16		32		48		64	72
9			18	27	36			63		

239.

×	3	4	5	6	7	8	9
4							
5							
6							
7							
8							
9							

240.

×	4	5	6	7	8	9
3						
4						
5						
6						
7						
8						
9						

241.

×	3	4	5	6	7	8	9
6							
7							
8							
9							

242.

×	6	7	8	9
3				
4				
5				
6				
7				
8				
9				

243.

×	5	6	7	8	9
5					
6					
7					
8					
9					

244.

×	6	7	8	9
6				
7				
8				
9				

In the following exercises, multiply.

245. $0 \cdot 15$

246. $0 \cdot 41$

247. $(99)0$

248. $(77)0$

249. $1 \cdot 43$

250. $1 \cdot 34$

251. $(28)1$

252. $(65)1$

253. $1(240{,}055)$

254. $1(189{,}206)$

255. ⓐ $7 \cdot 6$ ⓑ $6 \cdot 7$

256. ⓐ 8×9 ⓑ 9×8

257. $(79)(5)$

258. $(58)(4)$

259. $275 \cdot 6$

260. $638 \cdot 5$

261. $3{,}421 \times 7$

262. $9{,}143 \times 3$

263. $52(38)$

264. $37(45)$

265. $96 \cdot 73$

266. $89 \cdot 56$

267. 27×85

268. 53×98

269. $23 \cdot 10$

270. $19 \cdot 10$

271. $(100)(36)$

272. $(100)(25)$

273. $1{,}000(88)$

274. $1{,}000(46)$

275. $50 \times 1{,}000{,}000$

276. $30 \times 1{,}000{,}000$

277. 247×139

278. 156×328

279. $586(721)$

280. $472(855)$

281. $915 \cdot 879$

282. $968 \cdot 926$

283. $(104)(256)$

284. $(103)(497)$

285. $348(705)$

286. $485(602)$

287. $2{,}719 \times 543$

288. $3{,}581 \times 724$

Translate Word Phrases to Math Notation

In the following exercises, translate and simplify.

289. the product of 18 and 33

290. the product of 15 and 22

291. fifty-one times sixty-seven

292. forty-eight times seventy-one

293. twice 249

294. twice 589

295. ten times three hundred seventy-five

296. ten times two hundred fifty-five

Mixed Practice

In the following exercises, simplify.

297. 38×37

298. 86×29

299. $415 - 267$

300. $341 - 285$

301. $6{,}251 + 4{,}749$

302. $3{,}816 + 8{,}184$

303. $(56)(204)$

304. $(77)(801)$

305. $947 \cdot 0$

306. $947 + 0$

307. $15{,}382 + 1$

308. $15{,}382 \cdot 1$

In the following exercises, translate and simplify.

309. the difference of 50 and 18

310. the difference of 90 and 66

311. twice 35

312. twice 140

313. 20 more than 980

314. 65 more than 325

315. the product of 12 and 875

316. the product of 15 and 905

317. subtract 74 from 89

318. subtract 45 from 99

319. the sum of 3,075 and 95

320. the sum of 6,308 and 724

321. 366 less than 814

322. 388 less than 925

Multiply Whole Numbers in Applications

In the following exercises, solve.

323. **Party supplies** Tim brought 9 six-packs of soda to a club party. How many cans of soda did Tim bring?

324. **Sewing** Kanisha is making a quilt. She bought 6 cards of buttons. Each card had four buttons on it. How many buttons did Kanisha buy?

325. **Field trip** Seven school busses let off their students in front of a museum in Washington, DC. Each school bus had 44 students. How many students were there?

326. **Gardening** Kathryn bought 8 flats of impatiens for her flower bed. Each flat has 24 flowers. How many flowers did Kathryn buy?

327. **Charity** Rey donated 15 twelve-packs of t-shirts to a homeless shelter. How many t-shirts did he donate?

328. **School** There are 28 classrooms at Anna C. Scott elementary school. Each classroom has 26 student desks. What is the total number of student desks?

329. **Recipe** Stephanie is making punch for a party. The recipe calls for twice as much fruit juice as club soda. If she uses 10 cups of club soda, how much fruit juice should she use?

330. **Gardening** Hiroko is putting in a vegetable garden. He wants to have twice as many lettuce plants as tomato plants. If he buys 12 tomato plants, how many lettuce plants should he get?

331. **Government** The United States Senate has twice as many senators as there are states in the United States. There are 50 states. How many senators are there in the United States Senate?

332. **Recipe** Andrea is making potato salad for a buffet luncheon. The recipe says the number of servings of potato salad will be twice the number of pounds of potatoes. If she buys 30 pounds of potatoes, how many servings of potato salad will there be?

333. **Painting** Jane is painting one wall of her living room. The wall is rectangular, 13 feet wide by 9 feet high. What is the area of the wall?

334. **Home décor** Shawnte bought a rug for the hall of her apartment. The rug is 3 feet wide by 18 feet long. What is the area of the rug?

335. **Room size** The meeting room in a senior center is rectangular, with length 42 feet and width 34 feet. What is the area of the meeting room?

336. **Gardening** June has a vegetable garden in her yard. The garden is rectangular, with length 23 feet and width 28 feet. What is the area of the garden?

337. **NCAA basketball** According to NCAA regulations, the dimensions of a rectangular basketball court must be 94 feet by 50 feet. What is the area of the basketball court?

338. **NCAA football** According to NCAA regulations, the dimensions of a rectangular football field must be 360 feet by 160 feet. What is the area of the football field?

Everyday Math

339. **Stock market** Javier owns 300 shares of stock in one company. On Tuesday, the stock price rose $\$12$ per share. How much money did Javier's portfolio gain?

340. **Salary** Carlton got a $\$200$ raise in each paycheck. He gets paid 24 times a year. How much higher is his new annual salary?

Writing Exercises

341. How confident do you feel about your knowledge of the multiplication facts? If you are not fully confident, what will you do to improve your skills?

342. How have you used models to help you learn the multiplication facts?